Lab Exercise 11- Deployments with Rolling Update and Recreate Strategies

Understand how to use the rolling update and recreate strategies for deploying applications using Kubernetes Deployments.

Step 1: Create a Deployment with Rolling Update Strategy

Create a YAML file for the deployment:

Create a file named **nginx-deployment-rolling.yaml** with the following content:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-rolling
spec:
  replicas: 10
  minReadySeconds: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 5
  template:
    metadata:
      labels:
```

```
    app: web
  spec:
   containers:
   - name: nginx
    image: hkshitesh/kubedemo:1.0
    ports:
    - containerPort: 80
```

```
Desktop — vim nginx-deployment-rolling.yaml — 102×57

apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-rolling
spec:
  replicas: 10
  minReadySeconds: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 5
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80

~
~
```

Apply the deployment:

```
kubectl apply -f nginx-deployment-rolling.yaml ; watch "kubectl get rs -o wide"
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f nginx-deployment-rolling.yaml ; kubectl get rs -
o wide -w
deployment.apps/nginx-deployment-rolling unchanged
NAME                                   DESIRED   CURRENT   READY   AGE   CONTAINERS   IMAGES
        SELECTOR
nginx-deployment-rolling-85cf8767cc   10        10        10      73s   nginx        hkshitesh/kubedem
o:1.0   app=web,pod-template-hash=85cf8767cc
```

Verify the deployment:

```
kubectl get deployments
kubectl get pods -l app=web
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get deployments
NAME                       READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment-rolling   10/10   10           10          117s
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -l app=web
NAME                                        READY   STATUS    RESTARTS   AGE
nginx-deployment-rolling-85cf8767cc-8jcm5   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-bntm6   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-cqkvf   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-dkg72   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-fkd92   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-h7668   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-ljwd4   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-sfqmv   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-t4mst   1/1     Running   0          2m11s
nginx-deployment-rolling-85cf8767cc-v75mc   1/1     Running   0          2m11s
mohdanas@Mohds-MacBook-Air Desktop %
```

Update the deployment to a new image:

```
kubectl set image deployment/nginx-deployment-rolling nginx=  hkshitesh/kubedemo:2.0
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl set image deployment/nginx-deployment-rolling nginx=hkshi
tesh/kubedemo:2.0
deployment.apps/nginx-deployment-rolling image updated
mohdanas@Mohds-MacBook-Air Desktop %
```

Monitor the rolling update:

```
kubectl rollout status deployment nginx-deployment-rolling
```

```
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 1 old replicas are pending termin
ation...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 1 old replicas are pending termin
ation...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 1 old replicas are pending termin
ation...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 of 10 updated replicas are avai
lable...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 of 10 updated replicas are avai
lable...
^[deployment "nginx-deployment-rolling" successfully rolled out
```

Verify the updated pods:

```
kubectl get pods -l app=web -o wide
```

```
[mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -l app=web -o wide                          ]

NAME                                      READY   STATUS    RESTARTS   AGE   IP            NODE
  NOMINATED NODE   READINESS GATES
nginx-deployment-rolling-8587c95656-25cd4   1/1     Running   0          67s   10.244.0.22   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-9jtqn   1/1     Running   0          67s   10.244.0.25   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-d6lb8   1/1     Running   0          67s   10.244.0.26   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-ddfkc   1/1     Running   0          42s   10.244.0.28   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-dld66   1/1     Running   0          46s   10.244.0.27   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-mw7pv   1/1     Running   0          67s   10.244.0.21   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-mwbzt   1/1     Running   0          67s   10.244.0.24   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-q4sd5   1/1     Running   0          40s   10.244.0.29   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-qwkmm   1/1     Running   0          67s   10.244.0.23   minikube
  <none>           <none>
nginx-deployment-rolling-8587c95656-ztjms   1/1     Running   0          37s   10.244.0.30   minikube
  <none>           <none>
mohdanas@Mohds-MacBook-Air Desktop %
```

Step 2: Create a Deployment with Recreate Strategy

Create a YAML file for the deployment:

Create a file named **nginx-deployment-recreate.yaml** with the following content:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-recreate
spec:
  replicas: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: nginx:  hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80
```

```
Desktop — vim nginx-deployment-recreate.yaml — 102×57

apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-recreate
spec:
  replicas: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80
```

Apply the deployment:

```
kubectl apply -f nginx-deployment-recreate.yaml ; watch "kubectl get rs -o wide"
```

```
[mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f nginx-deployment-recreate.yaml ; kubectl get rs ]
-o wide -w
deployment.apps/nginx-deployment-recreate unchanged
NAME                                   DESIRED   CURRENT   READY   AGE     CONTAINERS   IMAGES
          SELECTOR
nginx-deployment-recreate-85cf8767cc   10        10        10      58s     nginx        hkshitesh/kube
demo:1.0   app=web,pod-template-hash=85cf8767cc
nginx-deployment-rolling-8587c95656    10        10        10      7m14s   nginx        hkshitesh/kube
demo:2.0   app=web,pod-template-hash=8587c95656
nginx-deployment-rolling-85cf8767cc    0         0         0       10m     nginx        hkshitesh/kube
demo:1.0   app=web,pod-template-hash=85cf8767cc
```

Verify the deployment:

```
kubectl get deployments
kubectl get pods -l app=nginx-recreate
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get deployments

NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment-recreate   10/10   10           10          86s
nginx-deployment-rolling    10/10   10           10          11m
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -l app=web
NAME                                          READY   STATUS    RESTARTS   AGE
nginx-deployment-recreate-85cf8767cc-22tdz    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-46xm9    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-8cg5c    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-gqmg7    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-nvlkg    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-rxbcc    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-t6wnx    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-tj5pd    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-vqmdr    1/1     Running   0          2m26s
nginx-deployment-recreate-85cf8767cc-w85dd    1/1     Running   0          2m26s
nginx-deployment-rolling-8587c95656-25cd4     1/1     Running   0          8m42s
nginx-deployment-rolling-8587c95656-9jtqn     1/1     Running   0          8m42s
nginx-deployment-rolling-8587c95656-d6lb8     1/1     Running   0          8m42s
nginx-deployment-rolling-8587c95656-ddfkc     1/1     Running   0          8m17s
nginx-deployment-rolling-8587c95656-dld66     1/1     Running   0          8m21s
nginx-deployment-rolling-8587c95656-mw7pv     1/1     Running   0          8m42s
nginx-deployment-rolling-8587c95656-mwbzt     1/1     Running   0          8m42s
nginx-deployment-rolling-8587c95656-q4sd5     1/1     Running   0          8m15s
nginx-deployment-rolling-8587c95656-qwkmm     1/1     Running   0          8m42s
nginx-deployment-rolling-8587c95656-ztjms     1/1     Running   0          8m12s
```

Update the deployment to a new image:

```
kubectl set image deployment/nginx-deployment-recreate nginx=nginx:1.21.1
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl set image deployment/nginx-deployment-recreate nginx=ngin
x:1.21.1
deployment.apps/nginx-deployment-recreate image updated
```

Monitor the update:

```
kubectl rollout status deployment nginx-deployment-recreate
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl rollout status deployment nginx-deployment-recreate

Waiting for deployment "nginx-deployment-recreate" rollout to finish: 0 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 1 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 2 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 3 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 4 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 5 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 6 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 7 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 8 of 10 updated replicas are ava
ilable...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 9 of 10 updated replicas are ava
ilable...
deployment "nginx-deployment-recreate" successfully rolled out
```

Verify the updated pods:

```
kubectl get pods -l app=nginx-recreate -o wide
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -l app=web -o wide
NAME                                        READY   STATUS    RESTARTS   AGE    IP            NODE
   NOMINATED NODE   READINESS GATES
nginx-deployment-recreate-8755c7cc9-8t8df   1/1     Running   0          2m4s   10.244.0.49   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-9lflj   1/1     Running   0          2m4s   10.244.0.47   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-ckqds   1/1     Running   0          2m4s   10.244.0.48   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-gbqzh   1/1     Running   0          2m4s   10.244.0.45   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-hq85m   1/1     Running   0          2m4s   10.244.0.43   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-nw6dw   1/1     Running   0          2m4s   10.244.0.46   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-qxvfg   1/1     Running   0          2m4s   10.244.0.41   minikube
   <none>           <none>
nginx-deployment-recreate-8755c7cc9-rfb54   1/1     Running   0          2m4s   10.244.0.50   minikube
```

Step 3: Clean Up

Delete the deployments:

```
kubectl delete deployment nginx-deployment-rolling
kubectl delete deployment nginx-deployment-recreate
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete deployment nginx-deployment-rolling
deployment.apps "nginx-deployment-rolling" deleted
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete deployment nginx-deployment-recreate

deployment.apps "nginx-deployment-recreate" deleted
mohdanas@Mohds-MacBook-Air Desktop %
```

Verify that all resources are cleaned up:

```
kubectl get deployments
kubectl get pods -l app=nginx
kubectl get pods -l app=nginx-recreate
```

```
deployment.apps "nginx-deployment-recreate" deleted
mohdanas@Mohds-MacBook-Air Desktop % kubectl get deployments

No resources found in default namespace.
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -l app=nginx

No resources found in default namespace.
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -l app=nginx-recreate
No resources found in default namespace.
mohdanas@Mohds-MacBook-Air Desktop %
```